Faculty of Electrical Engineering
Electronics Department
1st Mcil-/Semester 1/Algebra 1/Year: 2025-2026
Tutorial Sheet 2: Complex Numbers
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Exercise 1.
To express the following complex numbers in algebraic form:

$$z_1 = \frac{3+6i}{3-4i},\ z_2 = -\frac{2}{1-i\sqrt{3}},\ z_3 = \frac{2+5i}{1-i} + \frac{2-5i}{1+i},\ z_4 = \left(\frac{1+i}{2-i}\right)^2$$

Exercise 2.
Determine the modulus and argument of the following complex numbers:

$$z_1 = \sqrt{3}+i,\ z_2 = 1-i,\ z_3 = \frac{\sqrt{3}+i}{1-i},\ z_4 = (\sqrt{3}+i)(1-i),\ z_5 = 1+i\sqrt{3}.$$

To deduce the cartesian form of :$(2z_1)^3$, $(z_2)^6$, $\left(\frac{1+i\sqrt{3}}{1-i\sqrt{3}}\right)^{303}$, $\left(\frac{1+i\sqrt{3}}{2}\right)^{2010}$

Exercise 3.

1. *Determine the modulus and argument of* $u = 1+i$ *and* $v = 1+i\sqrt{3}$.
2. *Find, in two different ways (algebraic and trigonometric form), the two square roots of* u*, and then deduce the values of* $\cos(\frac{\pi}{8})$ *and* $\sin(\frac{\pi}{8})$*).*
3. *By writing* $\frac{u}{v}$ *in algebraic form, deduce the values of* $\cos(\frac{-\pi}{12})$ *and* $\sin(\frac{-\pi}{12})$*)*

Exercise 4. *Solve the equations in* $\mathbb{C}$:

1. $z^2+(1-2i)z+3+i=0$, $z^2+(1+2i)z+1+7i=0$, $(3+i)z^2-(8+6i)z+25+5i=0$
2. $z^3=-1$, $z^5=i$, $z^4=\frac{(1+i\sqrt{3})^4}{(1+i)^2}$
3. $(z-1)^5-(z+1)^5=0$

Exercise 5. *Solve in* $\mathbb{C}$ *the polynomial :*

- $z^3-(1+2i)z^2+3(1+i)z-10(1+i)=0$ *where knowing that it has a purely imaginary solution.* حل (iy)
- $4iz^3+2(1+3i)z^2-(5+4i)z+3(1-7i)=0$ *where knowing that it has a at least one real solution.* حل $(x.)$